

COMP1011 – Assignment 1

Problem: Basic Physics Calculator

Write a C++ program that performs one of four basic physics calculations based on the user's choice. The program should:

1. Display Student Information: Show your student ID and name.
2. Display a Menu: Present a menu with the following four operations:
 - a. Calculate kinetic energy
(Formula: $KE = 0.5 \times mass \times velocity^2$)
 - b. Calculate potential energy
(Formula: $PE = mass \times gravity \times height$, use gravity = 9.81 m/s^2)
 - c. Calculate momentum
(Formula: $Momentum = mass \times velocity$)
 - d. Calculate acceleration
(Formula: $Acceleration = force / mass$)
3. User Input: Ask the user to select an operation by entering 1, 2, 3, or 4.
4. Get the relevant user inputs for the selected calculation.
5. Compute and print the result in a clear format, as shown in samples below

```
Student ID: 20250002
Student Name: Jamie Lee

Basic Physics Calculator
1. Calculate Kinetic Energy (KE = 0.5 x mass x velocity^2)
2. Calculate Potential Energy (PE = mass x gravity x height)
3. Calculate Momentum (Momentum = mass x velocity)
4. Calculate Acceleration (Acceleration = force / mass)
Enter your choice (1-4):
```

- **Input & output requirements**

Input one integer to choose an operation. Then input appropriate numbers for the selected calculation. The output should follow the format as shown in sample results.

- **Sample results**

Sample 1	Sample 2
<pre>Student ID: 20250002 Student Name: Jamie Lee Basic Physics Calculator 1. Calculate Kinetic Energy (KE = 0.5 x mass x velocity^2) 2. Calculate Potential Energy (PE = mass x gravity x height) 3. Calculate Momentum (Momentum = mass x velocity) 4. Calculate Acceleration (Acceleration = force / mass) Enter your choice (1-4): 1 Enter the mass (in kg): 10 Enter the velocity (in m/s): 5 The kinetic energy is: 125 Joules</pre>	<pre>Student ID: 20250002 Student Name: Jamie Lee Basic Physics Calculator 1. Calculate Kinetic Energy (KE = 0.5 x mass x velocity^2) 2. Calculate Potential Energy (PE = mass x gravity x height) 3. Calculate Momentum (Momentum = mass x velocity) 4. Calculate Acceleration (Acceleration = force / mass) Enter your choice (1-4): 2 Enter the mass (in kg): 10 Enter the height (in m): 5 The potential energy is: 490.5 Joules</pre>